

Mid-Evaluation: Pronoun Ambiguity Resolution in Language Models

Preliminary results on frozen WinoGrande-style evaluation

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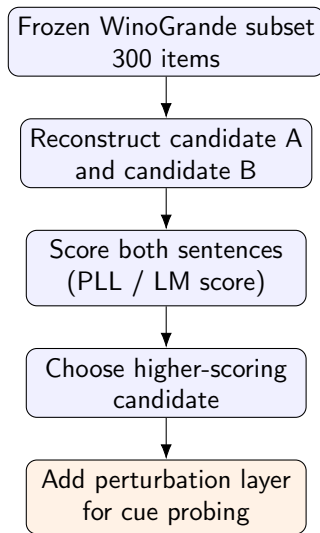
Question and Setup

Core question

- Do language models resolve Winograd/WinoGrande-style pronouns using human-like linguistic cues? Or are predictions driven by shallower lexical/statistical associations?

Evaluation setup

- Same first 300 WinoGrande validation items across comparable runs.
- Each item: sentence with blank, two candidate referents, gold answer.
- Reconstruct two candidate sentences by filling the blank with option 1 vs. option 2.
- Prediction = whichever reconstruction gets the higher model score.
- Perturbations are add as separate diagnostic layer; baseline stays on untouched sentences.



Stage 1: Baselines on the Frozen 300

Model / source	Items	Correct	Accuracy
N-gram baseline (baseline_experiment_v2.ipynb)	300	146	48.67%
BERT baseline (outputs_baseline/baseline_summary.json)	300	150	50.00%

Takeaway

- Under this setup, BERT is slightly above the n-gram baseline.
- Baseline accuracy alone does not show human-like reasoning.
- Score-gap diagnostic (Additional observation) : mean absolute gap is **4.14** on correct BERT cases vs. **4.31** on incorrect cases, so some wrong predictions are not obviously low-confidence.

Why Move Beyond Baselines?

Diagnostic motivation

- Accuracy on raw items does not tell us which cues drive the decision.
- We therefore probe cue sensitivity by changing the sentence while keeping the evaluation format fixed.
- Important separation: baseline results are always reported on untouched sentences.

Cue categories across BERT and RoBERTa

Category	What it probes
Semantic reasoning (BERT)	plausibility, causal fit, thematic fit
Lexical / surface (BERT)	wording swaps, lexical associations
Structural / discourse (BERT)	clause order, framing, accessibility
Synonym (RoBERTa)	lexical / surface substitution probe
Polarity (RoBERTa)	adjective flip; may also alter gold answer

Example Perturbations We Actually Used

Model / type	Cue label	Original	Perturbed
BERT	Semantic reasoning	"Terry tried to bake the eggplant in the toaster oven but the _ was too big." <i>Flip on balanced run: correct → incorrect</i>	"...but the _ was too large to fit."
BERT	Lexical / surface	"Judy bought new brushes ... The _ was too narrow." <i>Flip on balanced run: correct → incorrect</i>	"...The _ was too slim."
BERT	Structural / discourse	"Joel researched laws and helped to open a preschool for Eric. Because _ is very good with kids." <i>Flip on balanced run: incorrect → correct</i>	"Because _ is very good with kids, Joel researched laws and helped to open a preschool for Eric."
RoBERTa	Synonym (lexical / surface)	"The home that my parents had ... because the _ was sophisticated." <i>Flip in notebook output: correct → incorrect</i>	"The place that my parents had ... because the _ was sophisticated."
RoBERTa	Polarity	"James panicked ... but the _ is strong." <i>Flip in notebook output: correct → incorrect</i>	"James panicked ... but the _ is weak."

Stage 2: BERT Perturbation Analysis

Run	Category	n	Flip rate	Orig. → Pert. acc.
Pilot	Semantic reasoning	3	33.33%	66.67% → 33.33%
Pilot	Lexical / surface	3	0.00%	mixed single-item results
Pilot	Structural / discourse	3	0.00%	
Expanded	Lexical	6	33.33%	66.67% → 66.67%
Expanded	Structural	119	38.66%	51.26% → 44.54%
Balanced	Lexical / surface	20	5.00%	70.00% → 65.00%
Balanced	Semantic reasoning	20	5.00%	60.00% → 55.00%
Balanced	Structural / discourse	20	15.00%	55.00% → 70.00%

What this slide supports

- The **pilot** shows the method is capable of finding meaningful flips, but it is too small for category-level claims.
- The **expanded run** suggests BERT is quite sensitive to structural/discourse edits, but that run is heavily structural, so it is best read as a stress test.
- The **balanced 60-item run** is the cleanest comparison: lexical and semantic slices each show only 5% flip rates, while structural/discourse reaches 15%.
- Structural movement is not uniformly harmful: in the balanced run, structural reordering improves accuracy from 55% to 70%, so the effect is on *decision sensitivity*, not simply error.

Stage 3: RoBERTa Refinement

Model / operation	Items	Baseline	Perturbed	Flip rate
RoBERTa-large baseline ¹	300	58.00%	—	—
RoBERTa synonym	300	58.00%	56.00%	12.00%
RoBERTa polarity	36	58.33%	47.22%	11.11%

Conceptual mapping

- synonym → lexical / surface cue probe
- polarity → lexical / surface or semantic, depending on whether the gold referent is preserved

Preliminary comparison

- RoBERTa is stronger than BERT on the same 300-item baseline (58.00% vs. 50.00%).
- It still shows non-trivial sensitivity to synonym and polarity edits.

Current Interpretation

- Baseline gains over the n-gram reference are modest under the frozen 300-item setup.
- Perturbation analysis is more informative than raw accuracy because it reveals which cues can move the model's choice.
- BERT results currently suggest especially strong sensitivity to structural/discourse manipulations, but the effect direction varies by slice.
- RoBERTa improves baseline accuracy, yet synonym and polarity edits still shift predictions, which suggests remaining reliance on surface-sensitive cues.
- This perturbation setup establishes a robust framework for evaluating model reasoning, revealing initial insights into the gap between statistical guessing and true human-like comprehension.

Limitations and Next Steps

Current limitations

- Manual validation of the perturbations to confirm its meaningfulness.
- Synonym generation can create invalid substitutions unless curated. This curation again needs manual verifications.
- Due to the above constraints, sample size is 300 sentences (actual dataset extremely vast to compute).

Next steps

- Categorize perturbations consistently across BERT and RoBERTa.
- Systematically compare n-gram, BERT, and RoBERTa on matched item sets with supplementary literature review to derive insights.